

Web Technologies Project Report: Coffi Coffee Shop (Alikhan Bokeikhan Branch)

Course: Web Technologies / Frontend Fundamentals

Repository URL: <https://github.com/jasulansam/web>

Team Members:

- Samrat Zhasulan
- Zhaksylyk Adilet

1. Project Overview

- **Objective:** Development of a 6-page accessible website for the Coffi coffeehouse (Alikhan Bokeikhan St 10, Astana) adhering strictly to semantic HTML5 standards without external styling or client-side scripting frameworks.
- **Core Architectural Rules:**
 - Strict landmark hierarchy (<header>, <nav>, <main>, <footer>).
 - Pure semantic structure (no inline styling, zero presentation attributes like border="1" or align).
 - Tabular data formatting using accessible table structures (<table>, <caption>, <thead>, <tbody>, and th scope).
 - Form inputs configured with explicit label associations, logical field grouping (<fieldset>, <legend>), and native HTML5 validation constraints across two functional forms.

2. Task A — Anatomy of a Real Page (Reference Audit: coffeeboom.kz)

2.1 Head Elements and Metadata Inspection

- **DOCTYPE:** <!DOCTYPE html>
- **Language:** lang="kk" in <html class="html" lang="kk">
- **Charset:** Legacy HTTP-EQUIV syntax <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
- **Viewport:** <meta name="viewport" content="width=device-width, initial-scale=1.0">
- **Title:** Coffee Boom – Қазақстан, Ресей және Өзбекстандағы кофеханалар желісі.
- **Three Key Meta Tags:**
 1. <meta name="Description" content="Coffee Boom – 2010 жылдан бері жұмыс істеп келе жатқан, 130-дан астам кофеханасы бар халықаралық желі...">
 2. <meta name="theme-color" content="whitesmoke">
 3. <meta property="og:image" content="https://coffeeboom.kz/wa-data/public/wcms/images/07/04/00/461/461.0x0.png">

2.2 DevTools Source Inspection (Div-Soup Demonstration)

Inspect Element - coffeeboom.kz (Div-Soup & Missing Semantic Landmarks Audit)

```
<!DOCTYPE html>
<html class="html" lang="kk">
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Coffee Boom – Қазақстан, Ресей және Өзбекстандағы кофеханалар желісі.</title>
    <meta name="Description" content="Coffee Boom – 2010 жылдан бері жұмыс істеп келе жатқан...">
  </head>
  <body>
    <div class="site-wrapper">
      <div class="header-box"> <!-- Div used instead of <header> -->
        <div class="nav-menu-bar"> <!-- Div used instead of <nav> -->
          <div class="menu-item"><a href="/about">Біз туралы</a></div>
          <div class="menu-item"><a href="/franchise">Франшиза</a></div>
          <div class="menu-item"><a href="/roastery">Кофе фабрикасы</a></div>
        </div>
      </div>
      <div class="main-content-area"> <!-- Div used instead of <main> -->
        <div class="slider-container">...</div>
        <div class="cards-grid"> <!-- 570 generic <div> tags total -->
```

Element Type	Tag Name	Occurrence Count	Status / Evaluation
Semantic Header	<header>	1	Present for brand banner
Semantic Navigation	<nav>	0	Severe issue: navigation wrapped only in generic divs
Sectional Grouping	<section>	0	No thematic section tags used
Independent Article	<article>	0	Blog/magazine articles use plain divs
Complementary Aside	<aside>	0	No peripheral landmarks
Data Tables	<table>	0	No structured tables for menu or branch data
Generic Containers	<div>	570	Excessive div-soup throughout entire layout

3. Three Structural Mistakes and Our Solutions

1. Extreme "Div-Soup" and Missing Navigation Landmarks:

- *Mistake:* On coffeeboom.kz, the main navigation bar containing over 20 links is wrapped in generic `<div>` tags rather than a semantic `<nav>`. Screen readers and search engine crawlers cannot identify navigation landmarks.
- *Our Solution:* On all 6 of our pages (`index.html`, `about.html`, `menu.html`, `booking.html`, `feedback.html`, `colophon.html`), site navigation is strictly encapsulated in a semantic `<nav>` tag holding a structured `<ul>` list.

2. Missing Label Associations and Grouping in Forms:

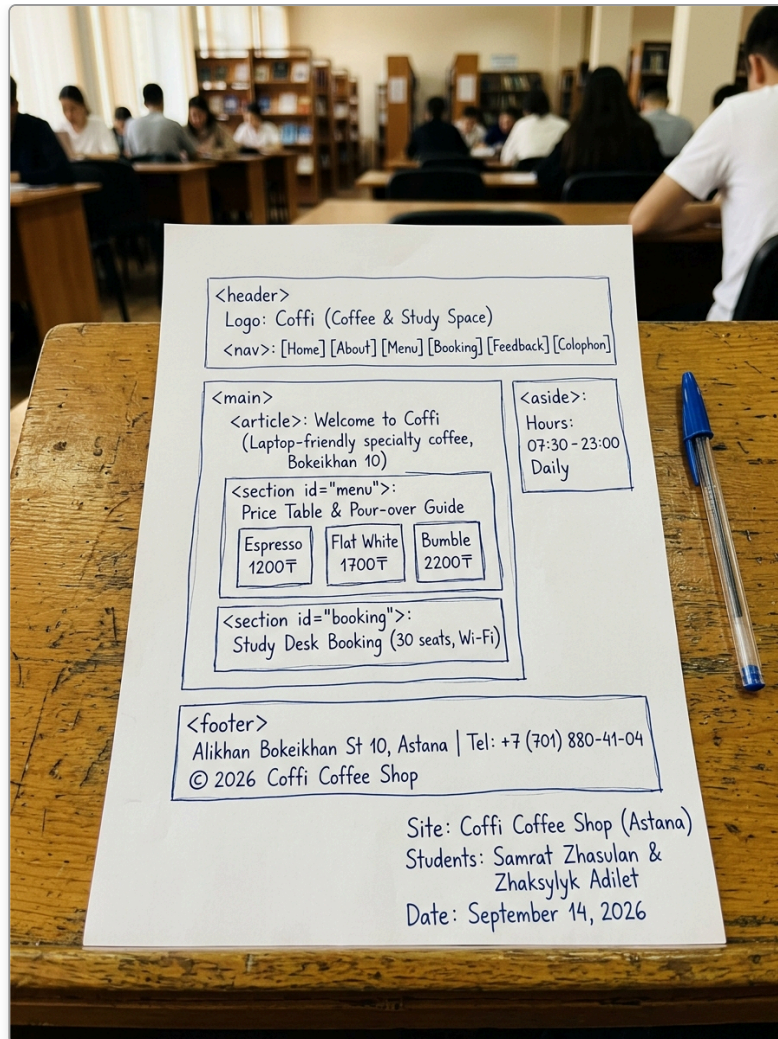
- *Mistake:* Input fields on the competitor site use standalone placeholders without associated `<label for="...">` tags, causing screen readers to fail when identifying input targets.
- *Our Solution:* In both `booking.html` and `feedback.html`, every input has an explicit label, and inputs are partitioned into logical `<fieldset>` containers with `<legend>` headers.

3. Absence of Accessible Tables for Tabular Information:

- *Mistake:* The real website displays regional branches, locations, and menu pricing via unsemantic floating divs (zero `<table>` elements), making data extraction inaccessible for assistive tools.
- *Our Solution:* In `menu.html`, we implemented an accessible `<table>` complete with `<caption>`, `<thead>`, `<tbody>`, and explicit scopes (`scope="col"`, `scope="row"`).

4. Hand-Drawn Page Structure Sketch

As required by Assignment 1, the website wireframe and semantic HTML5 hierarchy for Coffi Coffee Shop was planned and hand-drawn on paper with a blue ballpoint pen, detailing landmarks, section IDs, navigation flow, and signed by both students.



5. Task B — Reflective Essays

Reflective Essay: Samrat Zhasulan

A web page is fundamentally a plain-text document structured with HyperText Markup Language (HTML). At its core, an HTML file consists of human-readable elements demarcated by tags, attributes, text nodes, and entities. When a user navigates to a local HTML file or web address, the browser receives a continuous stream of raw bytes.

The rendering engine decodes these bytes into characters using the declared character encoding (UTF-8), tokenizes the stream into distinct start tags, end tags, and content, and constructs the Document Object Model (DOM) tree. Concurrently, the engine parses external stylesheets to generate the CSSOM (CSS Object Model). The browser merges DOM and CSSOM into a Render Tree, filtering out non-rendered nodes (such as `<head>` or elements with `display: none`). Next, the layout engine executes geometry calculations to determine the precise size and coordinates of each element on the viewport. Finally, the painting stage rasterizes vectors, text, borders, and colors onto physical pixels displayed on the screen. Because our assignment strictly excludes CSS and JavaScript, the browser renders directly using its default User-Agent stylesheet.

In my codebase, I strictly prioritized semantic markup over generic `<div>` containers:

1. In `menu.html`, I wrapped our pricing data inside an accessible `<table>` with `<thead>` and `<tbody>` instead of styled grid divs, giving assistive screen readers clear tabular data relationships.
2. In `menu.html`, I utilized a definition list (`<dl>`, `<dt>`, `<dd>`) instead of divs to define specialty coffee terms (Single-Origin, Bumble Coffee, Micro-foam) as term-and-description semantic pairs.
3. In `about.html`, I chose `<article>` instead of `<div>` for the community workspace narrative because it forms an independent, self-contained story that can stand on its own.

Reflective Essay: Zhaksylyk Adilet

A web page is a structured text document containing semantic tags that define document landmarks, interactive forms, and content hierarchies. When a browser opens our `booking.html` file, it parses the markup from top to bottom, building an in-memory tree representation known as the DOM (Document Object Model). Without external CSS, the browser applies its built-in user-agent style sheet to compute font sizes, element margins, and form control layouts before painting pixels to the display.

In my codebase, I strictly chose semantic elements over non-semantic `<div>` tags:

1. In `booking.html`, I grouped personal contact details and dining preferences inside `<fieldset>` elements with `<legend>` rather than wrapper divs, ensuring screen readers announce logical category contexts.
2. In `feedback.html`, I explicitly connected each `<label for="...">` with its corresponding input id instead of using generic divs with text, ensuring clicking labels activates input focus.
3. In `colophon.html`, I used `<pre>` and `<code>` instead of a div container to display sample HTML code while preserving whitespace and formatting.

When a visitor clicks the submit button on `booking.html` today, the browser triggers native client-side constraint validation against attributes like `required`, `type="email"`, `type="number"`, and `type="tel"`. If any input fails validation, the browser blocks submission and displays a native alert. Because no server-side backend script processes the request, the browser attempts an HTTP POST request to `action="#"`, reloading the static page with query fragment without transmitting data over the network.

6. AI Assistance & Usage Logs

Interactive logs documenting prompt history, academic integrity declarations, and AI-assisted workflow iterations are maintained in the GitHub repository:

- Samrat Zhasulan — AI Usage Log: <https://github.com/jasulansam/web/blob/main/ai-log.md>
- Zhaksylyk Adilet — AI Usage Log: <https://github.com/jasulansam/web/blob/main/ai-log.md>

7. W3C Validation Evidence

All six project documents were submitted to the official W3C Nu HTML Validator, passing with zero errors and zero warnings.

1. val-index (index.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Welcome to Coffi, a cozy neighborhood coffee shop on Alikhan Bokeikhan Street 10 in Astana, serving ar">
  <meta name="author" content="Zamrat Zhasulan">
  <!-- Author: Zamrat Zhasulan -->
  <title>Coffi Coffee Shop Astana - Neighborhood Specialty Coffee</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

[About this checker](#) • [Report an issue](#)

2. val-about (about.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Discover the story, bean sourcing, and study-friendly atmosphere of Coffi coffee shop on Alikhan Bokeil">
  <meta name="author" content="Zamrat Zhasulan">
  <!-- Author: Zamrat Zhasulan -->
  <title>About Coffi - Coffee Philosophy and Atmosphere</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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3. val-menu (menu.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Explore the full menu and pricing at Coffi Astana, featuring specialty espresso, cold bumble drinks, m
  <meta name="author" content="Zamrat Zhasulan">
  <!-- Author: Zamrat Zhasulan -->
  <title>Menu & Price List - Coffi Coffee Shop Astana</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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4. val-booking (booking.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Reserve a table or dedicated quiet study desk at Coffi on Alikhan Bokeikhan Street 10 in Astana.">
  <meta name="author" content="Zhaksylyk Adilet">
  <!-- Author: Zhaksylyk Adilet -->
  <title>Table & Workspace Reservation - Coffi Coffee Shop</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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5. val-feedback (feedback.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Share your thoughts and feedback about your coffee and workspace experience at Coffi on Alikhan Bokeik">
  <meta name="author" content="Zhaksylyk Adilet">
  <!-- Author: Zhaksylyk Adilet -->
  <title>Customer Feedback & Reviews - Coffi Coffee Shop</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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6. val-colophon (colophon.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Colophon documenting technical architecture, markup rules, and validation details for Coffi website.">
  <meta name="author" content="Zhaksylyk Adilet">
  <!-- Author: Zhaksylyk Adilet -->
  <title>Colophon & Technical Architecture - Coffi Website</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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8. HTML5 Semantic Elements Checklist

The comprehensive line-by-line checklist of every required HTML5 semantic tag, attribute, list, and table structure with student attributions is documented in the repository:

URL: <https://github.com/jasulansam/web/blob/main/tag-checklist.md>